

Requirements Traceability Matrix

Student Club Event Ticketing and Budget Portal

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Each requirement is linked to its use case, logical component, implementation and verification. Test names are defined in 6-Testing/test_portal.py. A test link means the behaviour was exercised; it does not turn a local benchmark into production acceptance.

Requirement	Use case	Component	Implementation	Verification	Status
FR-001	UC02 Review budget	Event and Budget	Portal.submit; Portal.review; approvals table	TC03, TC04, TC11	Passed
FR-002	UC01 Request budget; create or edit proposal	Event and Budget	Portal.create_event; Portal.submit; budget_items	TC01, TC02, TC05, TC13, TC19	Passed
FR-003	UC03 Register for event; generate QR ticket	Ticket Service	Portal.register; Portal.token; ticket_svg	TC06, TC07, TC08, TC16, TC18	Passed
FR-004	UC04 Scan and validate ticket	Check-in Service	Portal.checkin; conditional ticket update	TC09, TC10, TC11, TC12, TC17	Passed; hardware scan not tested
FR-005	UC05 Track event status	Dashboard Service	Portal.dashboard; /api/status; dashboard.js	TC11, TC14; browser walkthrough	API passed; polling interval 3 seconds
NFR-001	UC04 Scan and validate ticket	Authentication; Check-in; SQLite	Role checks; HMAC; atomic update	TC09-TC12, TC15, TC17; benchmark.py	Security cases passed; see measured performance limits
NFR-002	All browser use cases	Web UI; application; SQLite	Responsive CSS; status polling; HTTP routes	500-user benchmark; mobile viewport review	Partial verification; real-device and browser-load acceptance pending